A BEOL Ferroelectric FET-based Computing Unit for Digital Computing-in-Memory

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***Abstract*—Ferroelectric FETs (FeFETs) based on BEOL metal-ferroelectric-metal (MFM) capacitors exhibit a large memory window, high write endurance, and strong process compatibility. They not only serve as a promising candidate for embedded non-volatile memory but also provide a competent platform for developing computing-in-memory (CIM) chips. While prior works have extensively investigated the memory properties of MFM-FeFETs and their analog CIM applications, their use in digital CIM (DCIM) remains relatively underexplored. In this work, n-type MFM-FeFET devices are experimentally characterized for model calibration. Building on these results, we propose lookup-table (LUT)-based computing units for DCIM using MFM-FeFETs. A design guideline for circuit-device co-design is introduced to balance trade-offs among signal margin, readout yield, and area efficiency.**

Keywords—Ferroelectric Transistor, Compute in memory, LUT

I. Introduction

Computing-in-memory (CIM) based on non-volatile memory (NVM) can minimize data movement by enabling on-chip data storage and in-situ computing within the memory array [1-8]. Recently, digital CIM (DCIM) designs have become increasingly prevalent [10], achieving high energy efficiency and computing accuracy. However, most established DCIM designs rely on volatile memory, whereas NVM-based DCIM approaches remain relatively underexplored.

Ferroelectric field-effect transistors (FeFETs) based on hafnium zinc oxide (HZO) have attracted significant attention as a candidate for next-generation non-volatile memory (NVM) [10][11]. Compared to other emerging memories (Fig. 1), such as resistive random-access memory (RRAM), phase-change memory (PCM), and magnetic random-access memory (MRAM), they exhibit low write energy, a large signal ratio, and competitive write endurance. Metal-ferroelectric-metal (MFM)-FeFETs are fabricated by integrating the ferroelectric capacitor in the back-end-of-line (BEOL) process. Compared to conventional FeFETs with the ferroelectric layer in the front-end-of-line (FEOL) process, they demonstrate improved write endurance by eliminating the SiOx interfacial layer. Furthermore, the MFM-FeFETs feature high logic compatibility. Considering these advantages, the MFM-FeFETs provide a promising technology platform for developing NVM-based DCIMs.

In this work, we propose look-up-table (LUT)-based computing units for DCIM using MFM-FeFETs. The n-type MFM-FeFET devices are firstly characterized in experiment, then a SPICE model is calibrated based on the experimental data. On this basis, the design-technology co-optimization are carried out, which results in a design guideline to balance trade-offs among signal margin, readout yield, and area efficiency.

	SRAM	RRAM	PCM	STT-MRAM	FeFET (Hf-based)	MFM-FeFET (Hf-based)
Cell Structure	6T	1T-1R	1T-1PCM	1T-1MTJ	1T	1T-1C
Non-volatility	No	Yes	Yes	Yes	Yes	Yes
Typical Signal ratio	Large	Medium	Medium	Small	Large	Large
Write Endurance	$>10^{15}$	10^3~10^7	$>10^6$	$>10^{15}$	10^5-10^9	10^6-10^9
Write Energy	Very Low	Medium	Medium	Medium	Low	Low
Standby Power	High	Low	Low	Low	Low	Low

Fig. 1. Comparision between different emerging NVM technologies.

II. Proposed MFM-FeFet LUT Units For DCIM

A. Overall Macro Structure

Fig. 2 shows the overall structure of the proposed MFM-FeFET-based DCIM macro. It consists of a 128×128 array of ferroelectric look-up table units ($FLU_{0\text{-}127,0\text{-}127}$), an input decoder and driver, MFM-FeFET program/erase (P/E) circuitry, and a digital shifter-and-adder (DSA) for processing multi-bit data. In a single clock (CLK) cycle, the FLU array performs multiply-and-accumulate (MAC) operations between 256 1-bit inputs ($IN_{0\text{-}255}$) and 256×128 1-bit weights ($W_{0\text{-}255,0\text{-}127}$). Each $FLU_{i,j}$ processes the MAC operation between two 1-bit inputs (IN_{2i} and IN_{2i+1}) and

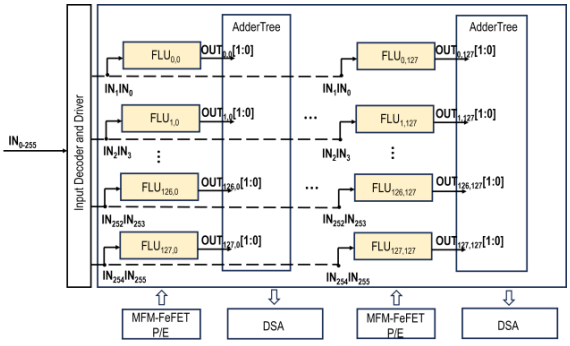
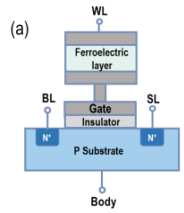


Fig.2. Proposed BEOL MFM FeFET-based DCIM macro.

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(b)

	WL(V)	SL(V)	BL(V)	Body(V)
Program	3.3	0	floating	0
Erase	0	3.3	floating	3.3
Compute	1.2	1.2	floating	0

Fig. 3. The structure and operaton of the BEOL MFM-FeFET devie. (a) The cross-sectional view. (b) The biasing table for program, erase, and compute operation.

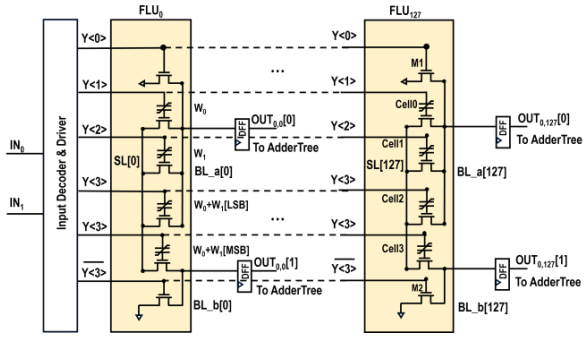


Fig. 4. The structure of the proposed FLU based on MFM-FeFETs.

two 1-bit weights ($W_{2i,j}$ and $W_{2i+1,j}$), generating a 2-bit output ($OUT_{i,j}$ [1:0]).

B. Proposed MFM-FeFET-based FLU

Fig. 3(a) shows the structure of the MFM-FeFET device used in this work. The MFM capacitor is integrated using BEOL process within the contact (CT) layer on top of the gate of a 180-nm core-voltage logic transistor. The area ratio (AR) is defined as the ratio between the area of the MFM capacitor and that of the transistor gate.

Fig. 3(b) shows the biasing table for program, erase, and compute operation. To program the cell to the low threshold voltage (V_{TH_L}) state, while the word-line (WL) is biased at 3.3V, the source-line (SL) and body are grounded. To erase the cell to the high threshold voltage (V_{TH_H}) state, while the SL and body are biased at 3.3V, the WL is biased at 3.3V. In the compute mode, while the body is grounded, the WL and SL are biasing at 1.2V. The state of the MFM-FeFET is readout using the source following approach. If the device is at the V_{TH_L} state, the bit-line (BL) voltage approximately equal to 1.2V – V_{TH_L}. If the device is at the V_{TH_H} state, the BL voltage approximately equal to 1.2V – V_{TH_H}.

Fig. 4 shows the detailed structure of the proposed FLU based on MFM-FeFETs. Each FLU consists of four MFM-FeFET cells ($Cell_{0-3}$) for weight data encoding, two logic n-type transistors (M1 and M2) as the pull-down cells, and two D-type flip-flops (DFFs) to latch the output data (OUT). The four MFM-FeFETs have a common SL and two split BLs (BL_a and BL_b).

TableI. The weight data encoding method in the proposed FLU.

W_1W_0	00	01	10	11
$Cell_0$	V_{TH_H}	V_{TH_L}	V_{TH_H}	V_{TH_L}
$Cell_1$	V_{TH_H}	V_{TH_H}	V_{TH_L}	V_{TH_L}
$Cell_2$	V_{TH_H}	V_{TH_L}	V_{TH_L}	V_{TH_H}
$Cell_3$	V_{TH_H}	V_{TH_H}	V_{TH_H}	V_{TH_L}

TableII. The computational turth table of the proposed FLU.

IN_1	IN_0	Y<3:0>	Select Cell	OUT[1]	OUT[0]
0	0	0001	M_2&M_1	0	0
0	1	0010	M_2&$Cell_0$	0	W_0
1	0	0100	M_2&$Cell_1$	0	W_1
1	1	1000	$Cell_3$&$Cell_2$	W_0+W_1 [MSB]	W_0+W_1 [LSB]

Each FLU performs the MAC operation between two 1-bit inputs (IN_0IN_1) and two 1-bit weights (W_0W_1). To perform CIM operations, the weight data are pre-loaded into the cell array and encoded by four cells. The $cell_0$ and $cell_1$ are programmed according to W_0 and W_1, respectively. The least-significant-bit (LSB) of W_0+W_1 (W_0+W_1[LSB]) is programmed into $Cell_2$. The most-significant-bit (MSB) of W_0+W_1 (W_0+W_1[MSB]) is programmed into $Cell_3$. Notice that the MFM-FeFET cell is programmed to V_{TH_L} to represent the weight-1, and the MFM-FeFET cell is programmed to V_{TH_H} to represent the weight-0. The weight data encoding method is shown in Table I.

The MFM-FeFET cells and pull-down transistors are selected via decoded control signals (Y<0-3>) derived from input bits IN_0 and IN_1. The selection logic operates as follows. When IN_0/IN_1 = 0/0, only Y<0> is activated, M_1 and M_2 are selected. The OUT is forced to 0, independent of the stored weight data. When IN_0/IN_1 = 1/0, only Y<1> is activated, $Cell_2$ and M2 are selected. The OUT reflects the value of W_0. When IN_0/IN_1 = 0/1, only Y<2> is activated, $Cell_3$ and M2 are selected. The OUT reflects the value of W_1. When IN_0/IN_1 = 1/1, Y<3> is activated, enabling $Cell_2$ and $Cell_3$. The OUT represents the sum of W_0 + W_1. The truth table in the computing mode is shown in Table II.

III. Result And Discussion

The test-key of the MFM-FeFET device is fabricated using a standard 180-nm logic process. The length (L) and width (W) of the logic transistor are 10000 nm and 180 nm, respectively. The length (L) and width (W) of the MFM capacitors are 1340 nm and 1340 nm, respectively. Fig. 5 shows the typical BL current (I_{BL}) as a function of the WL voltage (V_{WL}) measured from our devices after program and erase. The conditions for program and erase are listed in Fig. 3(b) and the AR of the device is 1. The measured threshold

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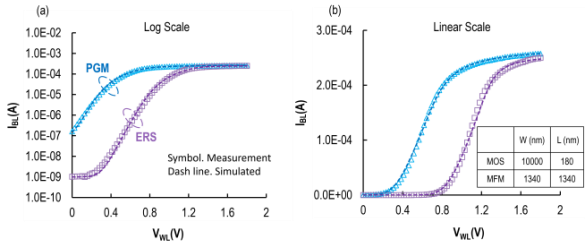


Fig. 5. Measured I_{BL} as a function of V_{WL} of the fabricated MFM FeFET device and the SPICE mode fitting results. (a) I_{BL} in linear scale. (b) I_{BL} in log scale.

voltage of the erased state is about 0.76 V. The measured threshold voltage of the programmed state is about 0.23 V, resulting in a clear memory window (MW) about 0.53 V. In order to perform circuit simulation based on the fabricated devices, we calibrate a SPICE model [12] based on the experimental data. From the I-V plots in linear (Fig. 5(a)) and logarithmic (Fig. 5(b)) scales, it can be seen that the model fits well with the experimental data in both the sub-threshold and linear regimes. This result validates our model calibration.

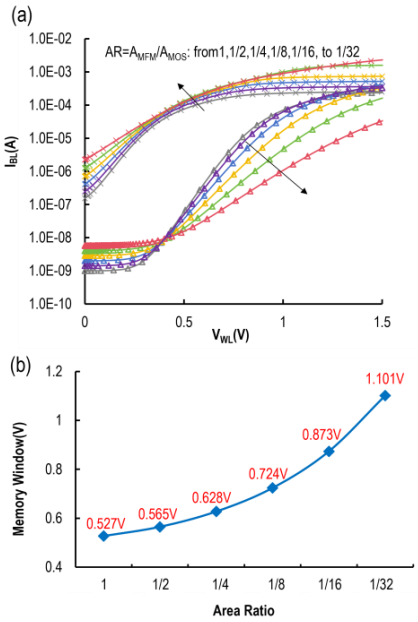


Fig.6. The impact of AR on the I-V properties of the MFM-FeFETs. (a) Simulated I-V curves at the programmed and erased states of the device with different ARs. (b) Dependence of MW on the AR.

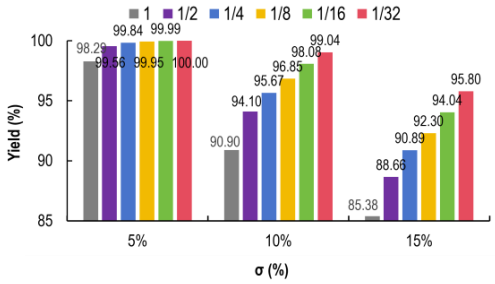


Fig. 8. The simulated computing yield as a function of device variations of the FLUs having MFM-FeFETs with different ARs.

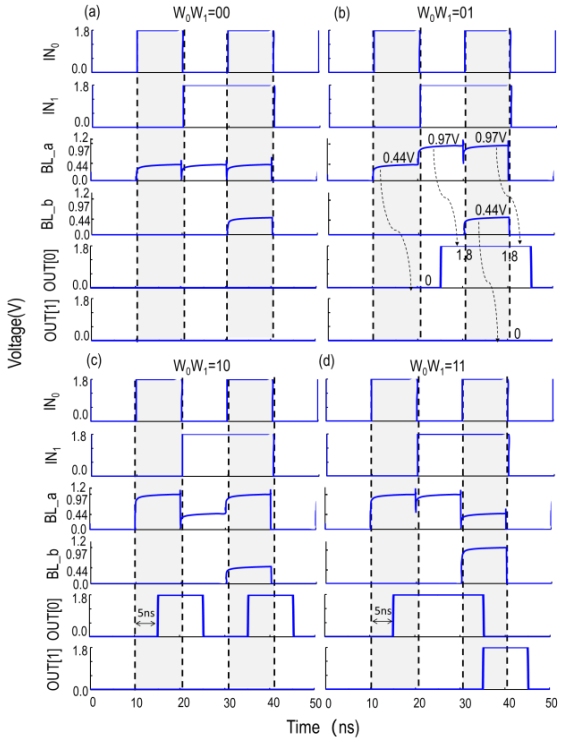


Fig. 7. Simulated waveforms of the FLU with different input and weight data pattens. (a) W_0=0, W_1=0. (b) W_0=0, W_1=1. (c) W_0=1, W_1=0. (d) W_0=1, W_1=1.

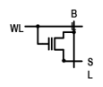
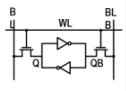
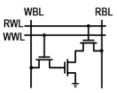
In Fig.6(a), we simulated the I-V properties of the MFM-FeFETs having different ARs. In this simulation, while the area of the MFM capacitor is fixed, the gate area of the transistor varies to achieve different ARs ranging from 1, 1/2, 1/4, 1/8, 1/16 to 1/32. The MW as a function of AR is further shown in Fig. 6(b). It can be seen that the MW increases with decreasing ARs. In particular, when the area ratio is reduced to 1/32, a large MW about 1.1V can be obtained. Notice that although reducing AR can improve MW, it results in increased bit-cell size, incurring a large degradation in the storage density.

Fig. 7 shows the simulated operational waveforms of the proposed FLU based on the MFM-FeFET devices having AR=1. The FLU functions are simulated across different data pattens in different cases of stored weights, including (a) W_0=0/W_1=0, (b) W_0=0/W_1=1, (c) W_0=1/W_1=0 and (d) W_0=1/W_1=1. Each waveform consists of four computing cycles with inputs varying from IN_0=0/IN_1=0, IN_0=1/IN_1=0, IN_0=0/ IN_1=1, and IN_0=1/ IN_1=1.The FLU is able to output all correct logic values in all situations, which justify its function.

To further address the variation issue, we analyze the computing yield of the proposed FLU under different variations. The computing yield is defined as the ratio of correct MAC operations to total MAC operations. As the standard deviation (σ) increases from 5% to 15%, the yield substantially degrades. When σ=15%, the yield remains below 98% regardless of the AR ratio. Although decreasing the AR ratio can improve MW and hence enhance computing yield, it incurs considerable density degradation.

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Table III Comparison with previous NVM-based DCIM

	NOR-FLASH-based LUT [13]	SRAM-based LUT [9]	eDRAM-based LUT [14]	FeFET-based LUT (This work)
Cell Structure				
Bit Cell	1T NOR Flash	6T SRAM	3T eDRAM	1 HZO MFM-FeFET
Memory Type	Non-Volatile	Volatile	Volatile	Non-Volatile
Readout Principle	Read through SA	Read through BL &BLB	Read through Refresh&Encode Port	Read through BL

Our evaluation shows that achieving a computing yield >98% with MFM-FeFET (AR=1) requires suppressing device variation below 5%. This result indicates the program-and-verify techniques [10] play in vital role to enable reliable MFM-FeFET DCIMs.

IV. Summary

Because of the low write power, high write endurance, and good logic compatibility, the MFM-FeFETs provide a promising technology platform for developing NVM DCIMs. In this work, we proposed the FLU based on the MFM-FeFETs for DCIM. The n-type MFM-FeFET test-keys are fabricated using a 180-nm standard logic process and experimentally tested. The measured I-V curves are well fitted using a SPICE model. On this basis, the functions of the proposed FLU are verified by circuit simulations. Furthermore, we have investigated the trade-offs among signal margin, readout yield, and area efficiency. It indicates that it is of importance to suppress the device variation below 5% to enable highly compact and reliable DCIM designs. Compared to previous DCIMs (Table III), this work firstly presents a DCIM design based on the MFM-FeFET cell array using the source following mechanism.

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